Why We Built V2

From a winner-oriented prototype to a validated debate-analysis framework

1. V1 demonstrated that an automated debate-judging pipeline can combine multiple signals and produce an overall speaker score.
2. But V1 tried to answer “Who won?” before establishing whether its measurements, constructs, weights, and ground truth were sufficiently valid.
3. V2 changes the research objective: first establish what can be measured reliably; then use those measurements to compare speakers.

What V1 Revealed

The main issue was measurement validity, not simply implementation

- Argument-quality mismatch: IBM ArgQ-Rank-30k uses short, standalone arguments and a fixed topic set; this does not directly match multi-sentence, dynamically generated debate turns.
- Manual weighting: heterogeneous metrics were combined with manually selected weights without enough empirical ground truth to justify their relative importance.
- Single-score information loss: one number can hide meaningful differences—for example, stronger stance support versus stronger rebuttal effectiveness.

Why Not Simply Improve V1?

The research problem is deeper than making the existing pipeline more robust

V1 starts with: “Given a debate, who is the winner?”

That question requires trustworthy individual measurements, justified aggregation/weights, and an appropriate ground truth.

During V1 development, those assumptions were not yet established.

V2 treats the task as a research-framework problem

Debate Judge V2

A multidimensional framework for measuring argumentative quality

Instead of one “winner” score, the system builds a profile of how each speaker argues.

1. What are we trying to measure?

Debate quality is treated as several related dimensions.

- How well does a speaker support their own position?
- Do their reasons actually support their claims?
- Do they remain consistent throughout the debate?
- Do they meaningfully answer and weaken the opponent?
- Do they introduce new information rather than repeat points?
- Which arguments survive after attacks and defenses?

2. The output: a speaker profile

The system explains strengths and weaknesses instead of collapsing everything into one number.

Dimension	Speaker A	Speaker B
Backs their side	High	High
Good reasoning	Medium	High
Stays consistent	High	Medium
Answers opponent	High	Low
Adds new info	Medium	High
Points survive	High	Medium

3. Common foundation

A small set of shared NLP components supports the metrics.

NLI model

How do two statements relate?

DeBERTa estimates agreement, unrelatedness, and disagreement. Many metrics use these probabilities.

LLM

What does the argument mean?

Ollama-based LLM calls extract claims/reasons and can identify reasoning flaws.

Argument Miner + Topic

What are the basic units?

Extracts claim/reasons and creates clear PRO/CON topic sentences before scoring.

- Elasticsearch is used for Delta and Votes to compare against similar real Reddit arguments.

4. Turn-level metrics

These ask what is happening inside an individual speaker turn.

Stance Support

Is this turn backing its own side?

Compare the turn with the PRO/CON topic hypothesis using NLI. Agreement with the speaker's side = support.

Logical Cogency

Do the reasons support the claim?

Combine acceptability, relevance, and sufficiency. NLI provides the core signals; careful mode can use LLM flaw detection.

Consistency

Is the speaker contradicting earlier claims?

Compare the current claim with earlier claims. Strong disagreement indicates an internal contradiction.

Delta

Does it resemble a mind-changing argument?

Similarity to real Change My View arguments that earned a Delta gives a bonus.

Votes

How well-received are similar arguments?

Use the top similar real arguments and compute a similarity-weighted quality/popularity score.

5. Logical Cogency — the core reasoning metric

It decomposes “good reasoning” into three interpretable questions.

Acceptability

Are the reasons believable?

Estimated as 1 − disagreement between reason and claim.

Relevance

Are the reasons related to the claim?

Estimated as 1 − unrelated probability.

Sufficiency

Are the reasons enough?

Fast mode uses agreement strength; careful mode can reduce the score when the LLM finds reasoning flaws.

- Current starting combination: 0.20 × Acceptability + 0.35 × Relevance + 0.45 × Sufficiency.
- These weights are research starting points and still need validation against human ratings.

6. Debate-level metrics

These look across multiple turns and the evolution of the argument.

Rebuttal Effectiveness

Did the speaker answer and weaken the opponent?

Three factors are combined: relevance × attack strength × quality of the rebuttal. The opponent's argument strength is then reduced by the resulting damage.

Information Density

Is the speaker adding new points or repeating?

New points are separated from statements already covered by earlier points. Both NLI and an LLM-based method are used side by side.

Argument Survival

Which arguments are still standing?

Arguments start at health 1.0, lose health when attacked, and regain some health when successfully defended.

7. How the whole system fits together

From raw debate text to an interpretable research profile.

1. **Read debate turns**
2. **Extract claims + reasons**
3. **Calculate turn-level metrics**
4. **Calculate debate-level metrics**
5. **Build per-speaker profile**
6. **Validate against human ratings**

- The final result is not “Speaker A = 73”. It is a multidimensional explanation of how each speaker performed.

9. Key limitations to keep in mind

The system is a measurement model, not a formal proof of argument quality.

- NLI-based agree/disagree signals can miss hidden logical gaps.
- Rebuttal damage and argument health are modeled quantities, not directly observed facts.
- New information is not automatically good information., need evidence

Debate Judge V2 — Further Work

From implemented metrics → tested and validated research system

Current focus: testing, improving validity, explaining metric decisions, and extending the debate representation.

1. Where we are now

The implementation exists; the next stage is research evaluation and refinement.

Implemented

The core metrics and their scoring logic have been implemented.

Testing started

A test HTML interface is being used to run individual metrics and record results.

Still experimental

NLI and LLM outputs must be tested across different argument types before we can judge their validity.

- At the moment, four metrics are being actively tested.
- The goal is not only to make the code run, but to check whether the scores make sense for real debate arguments.

2. Main next step: systematic metric testing

The testing phase is where the research assumptions can be challenged.

Test different argument types

Use varied debate turns rather than only straightforward examples:

- strong supporting arguments
- weak or unrelated arguments
- nuanced arguments
- rebuttals and counter-arguments
- arguments with multiple reasons
- potentially ambiguous cases

Record and inspect results

For each test, inspect:

- NLI probabilities
- LLM interpretation
- final metric score
- whether the result matches the expected behaviour
- cases where the model appears uncertain or wrong

Iterate and improve

When results reveal a problem:

1. Identify the failure case
2. Understand why the score behaved that way
3. Adjust the metric if justified
4. Test again on old + new examples

3. NLI + LLM: validation through testing

These models are the core evidence-generating components inside several metrics.

NLI — what we need to establish

NLI gives agreement / unrelated / disagreement probabilities.

The research question is not simply whether NLI runs correctly, but whether those probabilities behave sensibly for debate-specific comparisons.

We therefore need varied examples and expected outcomes.

LLM — what we need to establish

The LLM is used for harder interpretation tasks such as extracting argument structure or identifying reasoning issues.

We need to test whether its outputs are consistent and useful for the metric rather than accepting the output automatically.

- Result: testing provides evidence for whether the current metric design should be retained, adjusted, or extended.

4. Planned improvement: explain why a score is high or low

Move from a score-only output toward an evidence-based explanation.

Stance Support

If a turn receives a low stance-support score, identify the sentence(s) that appear unrelated to or inconsistent with the speaker's stance.

Positive evidence

If support is high, identify the sentence(s) that contribute most strongly to supporting the speaker's stance.

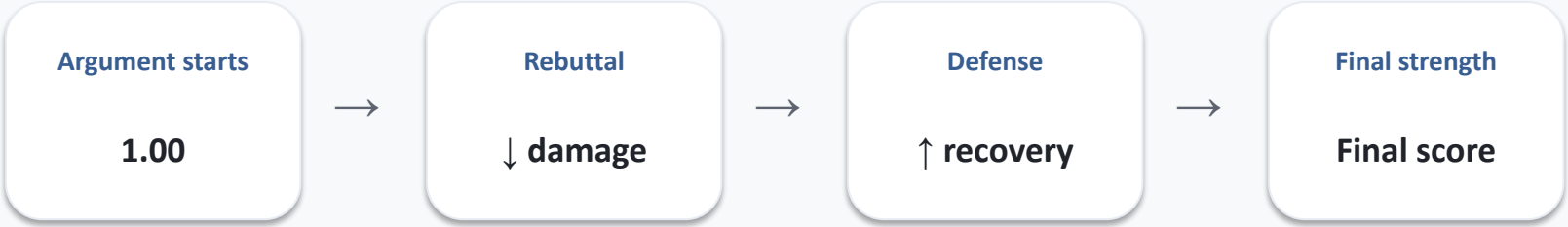
Why this matters

The professor/researcher can see not only “0.82 support”, but also the textual evidence behind that score.

This makes the metric more interpretable and easier to evaluate.

5. Argument Survival / Strength

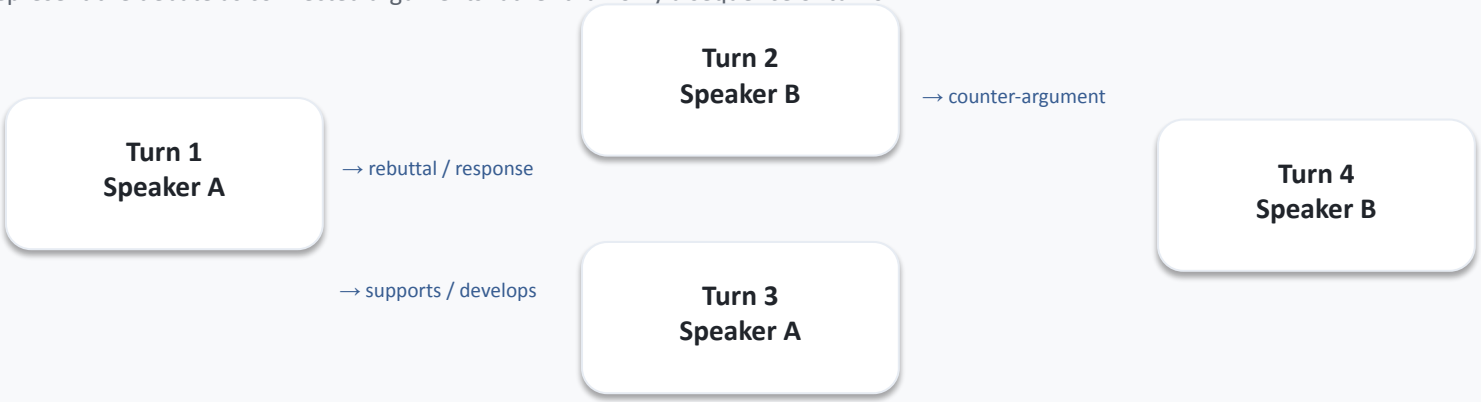
Track how an argument changes throughout the debate.



- Each speaker turn / argument begins with strength 1.0.
- A successful rebuttal decreases the argument's remaining strength.
- A valid defense can recover part of the lost strength.
- At the end of the debate, the remaining score represents how much of the argument survived.
- The current implementation exists; the next task is to test how efficiently and realistically it behaves.

6. Planned argument graph

Represent the debate as connected arguments rather than only a sequence of turns.



- Future goal: connect turns based on argumentative relationships.
- The graph could show which turn attacks which claim/reason and which arguments support or defend earlier arguments.
- This is planned work; it has not yet been implemented.

7. Testing the argument graph + survival together

A future view of the debate can combine structure and strength.

Graph structure

Who responds to whom?

Which claim is being attacked?

Is the response a rebuttal, counter-argument, or defense?

Argument strength

How strong was the argument before the attack?

How much damage did the rebuttal cause?

How much strength was recovered by defense?

Final research view

The graph can explain the path; the survival score can summarize the outcome.

Together they provide a more interpretable picture of argument evolution.

8. Planned testing loop

The system should evolve through evidence, not only implementation.

Test

Run metric on diverse arguments

Inspect

Compare output with expected behaviour

Find cases

Identify surprising / incorrect scores

Improve

Adjust logic, thresholds or model use

Retest

Check whether the change improves results

Document

Record what changed and why

9. Overall next-work roadmap

What we need to accomplish before treating the metrics as reliable.

NOW

- Continue testing the four active metrics
- Use varied argument types
- Record and inspect outputs
- Identify failure cases

NEXT

- Improve metric rules where justified
- Add sentence-level evidence for scores
- Test argument-survival behaviour
- Evaluate NLI / LLM consistency

THEN

- Implement argument graph
- Connect rebuttal / support relations
- Link graph structure with argument strength

RESEARCH VALIDATION

- Compare outputs with human judgments
- Measure agreement
- Refine the metrics based on evidence